

MICHIGAN DEPARTMENT OF ENVIRONMENTAL QUALITY

INTEROFFICE COMMUNICATION

TO: Sylvia Heaton, Permits Section, Water Resources Division

FROM: Phillip DePetro, Permits Section, Water Resources Division

DATE: April 12, 2018

SUBJECT: Cadillac WWTP
National Pollutant Discharge Elimination System (NPDES) Permit No. MI0020257
Water Quality-Based Effluent Limits (WQBELs)

Haring Twp WWTP
National Pollutant Discharge Elimination System (NPDES) Permit No. MI0059076
Water Quality-Based Effluent Limits (WQBELs)

Permits Section has evaluated conventional parameter water-quality based effluent limits (WQBELs) for the proposed reissuance of the NPDES Permits for the Cadillac WWTP (NPDES Permit No. MI0020257) and the Haring Twp WWTP (NPDES Permit No. MI0059076). The Cadillac WWTP is authorized to discharge 3.2 million gallons per day (MGD) of treated municipal wastewater to the Clam River at 44.2657°N -85.3985°E. The Haring Twp WWTP is authorized to discharge 0.135 MGD of treated municipal wastewater to the Clam River at 44.2805°N, -85.3865°E. The point of discharge for the Haring Twp WWTP is approximately 1.6 miles downstream of the Cadillac WWTP outfall. Both facilities are considered together in the WQBEL development process.

In the 1980s, WQBELs were developed based on a trout stream designation (coldwater protection) for the Clam River between Lake Cadillac and the Wexford-Missaukee county line in the non-summer seasons. This designation was modified in subsequent Department of Natural Resources Fisheries Orders to remove the trout stream designation for the entire year for that stretch of river, and therefore warmwater stream protection criteria apply in all months. This change results in less restrictive effluent limit recommendations in the fall, winter, and spring compared to current permit limits.

The Clam River from Lake Cadillac to the Wexford-Missaukee county line is protected for warmwater fish species, other indigenous aquatic life and wildlife, agriculture, navigation, industrial water supply, public water supply at the point of intake, partial body contact recreation, total body contact recreation from May 1 to October 31, and fish consumption. The Clam River downstream of the county line is protected for the same designated uses in addition to being protected for coldwater fish.

The monthly exceedance flows (in cfs) for the Clam River at the Cadillac WWTP outfall are as follows based on Low Flow Discharge Request 9152:

	<u>JAN.</u>	<u>FEB.</u>	<u>MAR.</u>	<u>APR.</u>	<u>MAY</u>	<u>JUNE</u>
50%	17	16	35	72	33	12
95%	5.4	4.8	8.6	13	6.7	4.0
	<u>JULY</u>	<u>AUG.</u>	<u>SEPT.</u>	<u>OCT.</u>	<u>NOV.</u>	<u>DEC.</u>
50%	7.2	6.1	6.5	11	18	19
95%	3.0	3.0	3.0	3.5	5.3	5.3

The 95% exceedance flow has increased in every month with the exception of December, where the flow has remained constant. The lowest monthly 95% exceedance flow in up to four seasons are used as the background flow of the receiving stream for the purposes of developing WQBELs. The following seasons are used, consistent with the current permit: May-September, October-November, December-March, and April.

A multiple reach Streeter-Phelps model was used to model the impact of both facilities on the Clam River at critical low-flow conditions. Effluent limit recommendations are determined for five-day carbonaceous biochemical oxygen demand (CBOD₅), ammonia (NH₃-N), and effluent dissolved oxygen (DO) that are protective of the DO standards at all times. Effluent limit recommendations are summarized in Table 1 for the Cadillac WWTP and Table 2 for the Haring WWTP.

The Clam River DO model has inputs based on many field-derived coefficients. Reaeration coefficients and the coefficients used to calculate river velocity were determined based on field study in 1983. The nitrification rate for the Clam River in the vicinity of the discharges was set to zero based on field studies and a previously calibrated/verified DO model. The diurnal variation between the daily average and daily minimum DO was set to 0.75 mg/l for the warmwater river reach and 0.51 mg/l for the coldwater river reach based on 2006 field study.

Effluent limit recommendations of the Cadillac WWTP in the summer season are less restrictive than current permit effluent limits. The current permit effluent limits are based on anti-degradation according to the last Basis of Decision Memo (7/8/2015). There is no daily maximum effluent limit recommendation for NH₃-N given that the nitrification rate has been set to zero.

Effluent limits reflective of federal secondary treatment standards (STS) are sufficient to protect the receiving water for the non-summer seasons at the Cadillac WWTP and all year at the Haring Twp WWTP. Based on department procedure, STS is conservatively assumed to include a daily maximum CBOD₅ concentration of 60 mg/l and a daily maximum NH₃-N concentration of 30 mg/l. A minimum effluent DO concentration of 3.0 mg/l is recommended to protect against acute toxicity in the effluent-receiving water mixing zone. No specific WQBEL recommendations are necessary for CBOD₅ and NH₃-N in seasons where STS-based effluent is protective. This is consistent with current permit effluent limits for the Haring Twp WWTP. The STS-based effluent limit recommendation is less restrictive than current permit effluent limits for the Cadillac WWTP; however, current permit effluent limits are based on the rationale of anti-degradation.

The discharges from each facility were also evaluated to determine 30-day average NH₃-N effluent limits that are protective of the chronic un-ionized NH₃-N toxicity criteria. The criterion is 0.053 mg/l of un-ionized NH₃-N for the warmwater section upstream of the county line and 0.029 mg/l of un-ionized NH₃-N for the coldwater section downstream. The facilities were considered together to ensure that the cumulative impact of the discharges did not result in chronic toxicity. Effluent limit recommendations for the Cadillac WWTP are given in Table 1, and are less restrictive than current permit effluent limits. No additional WQBEL limits are necessary for the Haring Twp WWTP discharge; the recommended effluent limits for this facility would be higher than the effluent quality expected from a plant producing effluent based on STS.

In the spring of 2017 the DEQ's Cadillac District asked if it would be possible to remove the month of May from the summer season of the Cadillac WWTP permit. Given Cadillac's northern location, it is possible that the cold weather season could extend into May. Cold temperatures

have the potential to limit ammonia removal via nitrification at the treatment facility and may cause effluent limit violations. Table 3 is provided to summarize what potential effluent limit recommendations would be under the following seasonal divisions: June-September, October-November, December-March, and April-May. Effluent quality based on STS is sufficient to protect the DO standard in the month of May in the updated model. The recommended 30-day average effluent NH₃-N limit of 5.2 mg/l in April-May would provide relief compared to the 0.9 mg/l monthly average limit currently in place.

These effluent limit recommendations are based on water quality standards. We have not addressed treatment practicality or cost effectiveness. Our recommendations do not imply that other considerations should not be taken into account when deciding on permit limits.

Table 1. Cadillac WWTP (MI0020257) WQBELs – Current Seasons

Parameter	Months	Conc. (mg/l)	Load (lb/d)	Basis	Rationale
CBOD ₅	May-September	17.0	454	Daily Maximum	DO Standard
NH ₃ -N	May-September	1.3	35	30 Day Average	Chronic Toxicity
	October-November	3.3	88	30 Day Average	Chronic Toxicity
	December-March	10.8	288	30 Day Average	Chronic Toxicity
	April	9.5	254	30 Day Average	Chronic Toxicity
DO	May-September	6.0	-	Minimum	DO Standard
	October-April	3.0	-	Minimum	Acute DO Toxicity

Design Flow = 3.2 MGD

Table 2. Haring Twp WWTP (MI0059076) WQBELs

Parameter	Months	Conc. (mg/l)	Load (lb/d)	Basis	Rationale
DO	All Year	3.0	-	Minimum	Acute DO Toxicity

Design Flow = 0.135 MGD

Table 3. Cadillac WWTP (MI0020257) WQBELs – Alternate Seasons

Parameter	Months	Conc. (mg/l)	Load (lb/d)	Basis	Rationale
CBOD ₅	June-September	17.0	454	Daily Maximum	DO Standard
NH ₃ -N	June-September	1.3	35	30 Day Average	Chronic Toxicity
	October-November	3.3	88	30 Day Average	Chronic Toxicity
	December-March	10.8	288	30 Day Average	Chronic Toxicity
	April-May	5.2	139	30 Day Average	Chronic Toxicity
DO	June-September	6.0	-	Minimum	DO Standard
	October-May	3.0	-	Minimum	Acute DO Toxicity

Design Flow = 3.2 MGD

cc: Brian Jankowski, Cadillac District Supervisor, Water Resources Division